

Non-Abelian gauge theories

1. Recap of gauge theory and brief look at Lie group theory

1.1 Recap: Abelian gauge theory

We firstly start with QED, which as we know is an Abelian $U(1)$ gauge theory. It start from the free fermion Lagrangian:

$$\mathcal{L} = i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi. \quad (3.1)$$

The Noether current is: $j^\mu = \bar{\psi}\gamma^\mu\psi$. To approach the QED Lagrangian, we introduce the (global) gauge transformation as:

$$\psi \rightarrow \psi' = e^{i\alpha}\psi \quad (3.2)$$

$$\bar{\psi} \rightarrow \bar{\psi}' = e^{-i\alpha}\bar{\psi} \quad , \quad \alpha \in \mathbb{R}. \quad (3.3)$$

The local gauge transformation is just changing the parameter α rely on the coordinate:

$$\alpha \rightarrow \alpha(x). \quad (3.4)$$

The gauge symmetry is saying that the Lagrangian is invariant under the gauge transformation. This requires us to introduce the covariant derivative D_μ :

$$D_\mu\psi(x) = (\partial_\mu - igA_\mu)\psi(x). \quad (3.5)$$

where in QED case, the charge $g = e$, and the gauge field A_μ is just the photon field, which satisfies the transformation rule:

$$A_\mu \rightarrow A'_\mu = A_\mu + \frac{1}{g}\partial_\mu\alpha(x). \quad (3.6)$$

Then we immediately have the covariant derivative transfers as

$$D_\mu\psi \rightarrow (D_\mu\psi)' = \partial_\mu\psi' - igA'_\mu\psi' = e^{i\alpha}D_\mu\psi. \quad (3.7)$$

Thus the invariant of $\mathcal{L}$ requires that

$$\mathcal{L}(\psi, A_\mu) = \mathcal{L}(\psi', A'_\mu) = i\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi \quad (3.8)$$

$$= i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi + gA_\mu j^\mu. \quad (3.9)$$

For photon dynamical, we need to add kinetic term for A_μ :

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \quad (3.10)$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

1.2 $SU(N)$ gauge theory

Now we consider the non-Abelian $SU(N)$ gauge theory with N Dirac spinors ψ_j , where $j = 1, \dots, N$. The original Lagrangian is

$$\mathcal{L} = \sum_{j=1}^N i\bar{\psi}_j \gamma^\mu \partial_\mu \psi_j - m\bar{\psi}_j \psi_j. \quad (3.11)$$

Under the global unitary transformation $U_{ij} \in \mathbb{C}$, $UU^\dagger = \mathbb{1}$, the spinors obey

$$\psi_j \rightarrow \psi'_j = \sum_k U_{jk} \psi_k, \quad (3.12)$$

$$\bar{\psi}_j \rightarrow \bar{\psi}'_j = \sum_k \bar{\psi}_k U_{kj}^\dagger, \quad (3.13)$$

thus the mass term transformation under $SU(N)$

$$\sum_j \bar{\psi}_j \psi_j \rightarrow \sum_j \bar{\psi}'_j \psi'_j = \sum_{k,l,j} \bar{\psi}_k \underbrace{U_{kj}^\dagger U_{jl}}_{\delta_{kl}} \psi_l = \sum_k \bar{\psi}_k \psi_k \quad (3.14)$$

is invariant.

Similar with the Abelian case, the local gauge transformation is $U \rightarrow U(x)$. We also require the Lagrangian invariant under the transformation. This gives coupling of fermions to gauge field in covariant derivative analogous to QED:

$$D_\mu \psi_j = \partial_\mu \psi_j - ig \sum_k (A_\mu)_{jk} \psi_k. \quad (3.15)$$

This form of covariant derivative also requires the transformation rule as

$$D_\mu \psi_j \rightarrow (D_\mu \psi_j)' = \partial_\mu \psi'_j - ig \sum_k (A'_\mu)_{jk} \psi'_k \quad (3.16)$$

$$= \sum_k U_{jk} (D_\mu \psi_k) \quad (3.17)$$

For convenient, we will use the short-hand notation(Matrix notation), which basically just ignore the index of the matrix element:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = U D_\mu \psi = e^{i\alpha(x)} D_\mu \psi. \quad (3.18)$$

We claim that the gauge field transforms following

$$A_\mu \rightarrow A'_\mu = U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger. \quad (3.19)$$

Although we can directly use the definition to compute this expression, I just don't want to write so much. Instead, we can have a quick check that this claim indeed satisfies our requirement:

$$D_\mu \psi \rightarrow (D_\mu \psi)' = \partial_\mu (U\psi) - ig \left(U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger \right) U\psi \quad (3.20)$$

$$= U \partial_\mu \psi + (\partial_\mu U) \psi - ig U A_\mu \psi - (\partial_\mu U) \psi \quad (3.21)$$

$$= U (\partial_\mu \psi - ig A_\mu \psi) = U D_\mu \psi. \quad (3.22)$$

Thus, the Lagrangian is invariant under $SU(N)$ gauge transformation.

Now we need to approach the dynamical of non-Abelian gauge field, which requires a field strength. Since we keep calling it as “non-Abelian”, but until now it’s not quite obvious to see how it is “non-Abelian”, therefore, the field strength should give you some direct idea when we claim:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]. \quad (3.23)$$

We can also check it is true that this field strength indeed satisfies our requirement, namely keep the Lagrangian invariant. To show this, we firstly have the transformation rule:

$$F_{\mu\nu} \rightarrow F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu - ig[A'_\mu, A'_\nu] = U F_{\mu\nu} U^\dagger. \quad (3.24)$$

The Lagrangian is proportion to the trace over $F_{\mu\nu} F^{\mu\nu} = (F_{\mu\nu})_{jk} (F^{\mu\nu})^{kj} = \text{tr}(F_{\mu\nu} F^{\mu\nu})$, due to the Lagrangian is a scalar, thus the only possible combination is the trace over $F_{\mu\nu}$. Thus $\text{tr} F_{\mu\nu}$ transfers as

$$\begin{aligned} \text{tr}(F_{\mu\nu} F^{\mu\nu}) &\rightarrow \text{tr}(F'_{\mu\nu} F'^{\mu\nu}) = \text{tr}(U F_{\mu\nu} U^\dagger U F^{\mu\nu} U^\dagger) \\ &= \text{tr}(U U^\dagger F_{\mu\nu} F^{\mu\nu}) \\ &= \text{tr}(F_{\mu\nu} F^{\mu\nu}). \end{aligned} \quad (3.25)$$

Then we get **the Lagrangian of Yang-Mills theory**

$$\mathcal{L} = -\kappa \text{tr}(F_{\mu\nu} F^{\mu\nu}) + \sum_{j,k=1}^N [\bar{\psi}_j \gamma^\mu (D_\mu)_{jk} \psi_k - m \bar{\psi}_j \psi_j], \quad (3.26)$$

where κ is normalized constant.

1.3 Recap for essential Lie group theory

To continue our journey, it’s necessary to master some basic knowledge about Lie group and Lie algebra. So in the next, we will recall some essential knowledge about these mathematical tools and show how are they used in gauge theory.

Firstly, we recall the infinitesimal transformation of the gauge operation. For Abelian theory, e.g. QED:

$$e^{i\alpha} \simeq 1 + i\alpha. \quad (3.27)$$

For non-Abelian gauge theory, e.g. Yang Mills theory:

$$U = \mathbb{1} + \hat{H} + \mathcal{O}(\hat{H}^2), \quad (3.28)$$

$$U^\dagger = \mathbb{1} + \hat{H}^\dagger + \mathcal{O}(\hat{H}^2). \quad (3.29)$$

Thus we have

$$U^{-1} = U^\dagger \implies \hat{H} = -\hat{H}^\dagger \quad (\text{anti-hermitian}) \quad (3.30)$$

where we note $\hat{H} = iH$ with $H = H^\dagger$. Expand $\hat{H}$ in parameters of transformation α^a

$$\hat{H}_{jk} = i \sum_a \alpha^a (t^a)_{jk} \quad (3.31)$$

with $\alpha^a \in \mathbb{R}$, $t^a = (t^a)^\dagger$ is hermitian. Thus t^a span a basis in the space of hermitian $N \times N$ matrices such that we can expand $\hat{H}$ in terms of this basis. t^a are called the generators of gauge transformations.

Generally, the unitary $N \times N$ matrix has N^2 linear independent generators due to t^a is $N \times N$. However, in our case, we require $\det U = 1$, which gives one constraint. So there are **$N^2 - 1$ generators in $\text{SU}(N)$** . Further more, base on $\det U = 1$ and the Jacobi identity $\det(e^M) = e^{\text{tr}(M)}$, we find

$$\det U = \det(e^{\hat{H}}) = e^{\text{tr} \hat{H}} = 1, \quad (3.32)$$

thus, $\text{tr} \hat{H} = 0$ and implies

$$\text{tr}(t^a) = 0. \quad (3.33)$$

So **t^a are traceless generators.**

Now we discuss the algebra structure for generators t^a . The Lie algebra, namely the commutator is given by:

$$[t^a, t^b] = i \sum_c f^{abc} t^c. \quad (3.34)$$

where f^{abc} is the structure constant of Lie algebra. The elements of Lie group can be expanded around $\mathbb{1}$.

$$g(\alpha) = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a + \frac{1}{2} \sum_{a,b=1}^d \alpha_a \alpha_b T^{ab} + \mathcal{O}(\alpha^3), \quad (3.35)$$

where $T^{ab} = T^{ba}$ is symmetric, which can be expanded in terms of the generator t^a, t^b . Notice $g(0) = \mathbb{1}$ and because of $g(\alpha)\mathbb{1} = \mathbb{1}g(\alpha)$, we then have

$$\gamma_a(\alpha_1, \dots, \alpha_d; 0, \dots, 0) = \gamma_a(0, \dots, 0; \alpha_1, \dots, \alpha_d) = \alpha_a \quad (3.36)$$

where the γ is defined by the group multiplication:

$$g(\alpha_1, \dots, \alpha_d) \cdot g(\beta_1, \dots, \beta_d) = g(\gamma_1, \dots, \gamma_d), \quad (3.37)$$

in which γ is the function of α and β :

$$\gamma_a = \gamma_a(\alpha_1, \dots, \alpha_d; \beta_1, \dots, \beta_d). \quad (3.38)$$

Base on the fact that $\gamma(\alpha, 0) = \alpha$ and $\gamma(0, \beta) = \beta$, then we have only one choice to expand γ_a until the second order to keep these relation, namely:

$$\gamma_a = \alpha_a + \beta_a + \sum_{b,c} C_{bc}^a \alpha_b \beta_c + \dots, \quad (3.39)$$

where C_{bc}^a is a unknown coefficient. Thus, use $g(\alpha)g(\beta) = g(\gamma)$, we have for the left-hand side:

$$g(\alpha)g(\beta) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + \frac{1}{2} \sum_{a,b} [(\alpha_a \alpha_b + \beta_a \beta_b) T^{ab} - 2\alpha^a \beta^b t^a t^b] + \dots, \quad (3.40)$$

And for the right-hand side:

$$g(\gamma) = \mathbb{1} + i \sum_a \gamma_a t^a + \frac{1}{2} \sum_{a,b} \gamma_a \gamma_b T^{ab} + \dots \quad (3.41)$$

Thus use (3.39), then the right-hand side becomes

$$g(\gamma(\alpha, \beta)) = \mathbb{1} + i \sum_a (\alpha_a + \beta_a) t^a + i \sum_{a,b,c} C_{bc}^a \alpha_b \beta_c t^a + \frac{1}{2} \sum_{a,b} (\alpha_a + \beta_a)(\alpha_b + \beta_b) T^{ab} + \dots \quad (3.42)$$

Compare (3.40) with (3.42), we find

$$-t^a t^b = i \sum_c C_c^{ab} t^c. \quad (3.43)$$

this gives the structure constant:

$$[t^a, t^b] = i \sum_{c=1}^d (C_c^{ba} - C_c^{ab}) t^c \quad (3.44)$$

$$= i \sum_{c=1}^d f^{abc} t^c \quad (3.45)$$

where $f^{abc} = -f^{bac}$, and d is the dimension of Lie algebra.

Examples of Lie groups

- **GL(N)**: Group of $N \times N$ matrices in $\mathbb{R}^N$ with $\det M \neq 0$.
- **U(N)**: Group of unitary $N \times N$ matrices in $\mathbb{C}^N$, satisfies

$$U^\dagger U = U U^\dagger = \mathbb{1} \quad \text{and} \quad \det U \neq 0. \quad (3.46)$$

Infinitesimal version:

$$U = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a. \quad (3.47)$$

with $d = N^2$ is the dimension of $U(N)$ Lie algebra/group.

- **SU(N)** := $\{A \in U(N) | \det A = 1\}$. Dimension of Lie algebra $\mathfrak{su}(N)$ is $d = N^2 - 1$. Generators t^a are traceless

$$\det U = \det \left(\mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \right) = 1 + i \sum_{a=1}^d \alpha_a \text{tr } t^a = 1 \quad (3.48)$$

thus

$$\text{tr } t^a = 0 \quad (3.49)$$

- **O(N)**: group of orthogonal $N \times N$ matrices in $\mathbb{R}^N$.

$$O O^T = O^T O = \mathbb{1} \quad (3.50)$$

Infinitesimal version:

$$O = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \quad (3.51)$$

Dimension of Lie algebra $\mathfrak{o}(N)$ is $\frac{1}{2}N(N-1)$. And notice:

$$O O^T = \left(\mathbb{1} + i \sum_{a=1}^d \alpha_a t^a \right) \left(\mathbb{1} + i \sum_{b=1}^d \alpha_b (t^b)^T \right) = \mathbb{1}, \quad (3.52)$$

thus,

$$\mathbb{1} = \mathbb{1} + i \sum_{a=1}^d \alpha_a t^a + i \sum_{b=1}^d \alpha_b (t^b)^T + \mathcal{O}(\alpha^2) \quad (3.53)$$

$$= \mathbb{1} + i \sum_{a=1}^d (\alpha_a t^a + \alpha_a (t^a)^T), \quad (3.54)$$

which implies

$$t^a = -(t^a)^T. \quad (3.55)$$

- $\mathbf{SO}(\mathbf{N}) := \{A \in O(N) \mid \det A = 1\}$
- $\mathbf{SO}(\mathbf{N}, \mathbf{M}) = \{\Lambda \in \text{Mat}(N+M, \mathbb{R}) \mid \Lambda^T \eta \Lambda = \eta, \det \Lambda = 1\}$, where

$$\eta = \text{diag}(\underbrace{1, \dots, 1}_N, \underbrace{-1, \dots, -1}_M) \quad (3.56)$$

- **Lorentz group:** $SO(1, 3)$

1.4 Representations of Lie algebra

Def: Representation of group

A **representation of a group G on a d_r -dimensional vector space V** is a group homomorphism

$$\rho : G \rightarrow GL(V), \quad (3.57)$$

which maps each group element $g \in G$ to an invertible linear operator $\rho(g)$ acting on V , such that:

$$\rho(g_i) \cdot \rho(g_j) = \rho(g_i g_j) \quad \forall g_i, g_j \in G. \quad (3.58)$$

The dimension d_r of the vector space V is called **the dimension of the representation**.

- A representation ρ is called **reducible** if there exists a non-trivial proper subspace $U \subset V$ (where $U \neq \{0\}$ and $U \neq V$) that is invariant under the action of all group elements, meaning:

$$\rho(g)u \in U \quad \forall g \in G, \forall u \in U. \quad (3.59)$$

- If no such invariant subspace exists, the representation is called irreducible (irrep).

For a Lie group characterized by continuous parameters α , a representation is a mapping $\rho : g(\alpha) \rightarrow M(\alpha)$, where the matrices (or operators) $M(\alpha)$ preserve the group multiplication:

$$M(\alpha) \cdot M(\beta) = M(\gamma) \quad \text{whenever} \quad g(\alpha) \cdot g(\beta) = g(\gamma). \quad (3.60)$$

In the next, we introduce some useful representations.

• Fundamental(defining) representation

For a matrix Lie group, the **fundamental representation** is simply the group of matrices themselves, acting on their natural vector space.

Example: $SU(2)$ fundamental representation

- **Representation Space (V):** A 2-dimensional complex vector space $\mathbb{C}^2$, physically corresponding to a spin-1/2 state

$$|\psi\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle. \quad (3.61)$$

- **Dimension of the representation:** $d = 2$ (since the matrices are 2×2). Note that the dimension of the $SU(2)$ group/algebra itself is $2^2 - 1 = 3$ (the number of

independent generators). The 3 Hermitian generators for this $d = 2$ fundamental representation are the Pauli matrices:

$$t^a = \frac{1}{2}\sigma^a \quad (a = 1, 2, 3), \quad (3.62)$$

which satisfy the Lie algebra $[t^a, t^b] = i\epsilon^{abc}t^c$, and the Pauli matrix identity:

$$\sigma^k \sigma^l = \delta^{kl} \mathbb{1} + i\epsilon^{klm} \sigma^m \quad (3.63)$$

Physically, we can use these fundamental building blocks to construct higher-dimensional representations via tensor products, such as the 3-dimensional spin-1 triplet state. In fact, $SU(2)$ has spin- j representations with dimension $2j + 1$. For instance, $j = 1/2$ corresponds to the 2-dimensional fundamental representation using 2×2 Pauli matrices.

- **Adjoint representation**

Instead of acting on physical states, the **adjoint representation** is a d -dimensional representation where the group/algebra acts on the Lie algebra space itself (spanned by the generators $\{t^a\}$ as basis vectors). The definition is as follows

Adjoint representation

We first define a automorphism map of group G :

$$\Psi : G \rightarrow \text{Aut}(G) \quad (3.64)$$

$$g \mapsto hgh^{-1} \equiv \Psi_h(g), \quad \forall h \in G, \quad (3.65)$$

where $\text{Aut}(G)$ is the automorphism group of G . Then we can define the derivative of Φ_g at the origin noted as Ad_g :

$$\text{Ad}_g = (d\Phi_g)_e : T_e G \rightarrow T_e G, \quad (3.66)$$

where $T_e G$ is the tangent space in the natural element of G . Thus the **adjoint representation** is a homomorphism:

$$\text{Ad} : G \rightarrow \text{Aut}(T_e G) \quad (3.67)$$

$$g \mapsto \text{Ad}_g. \quad (3.68)$$

To obtain the explicit matrix form of the adjoint representation, we consider an infinitesimal group element $g = e + i\alpha^a t^a \in G$. Acting on a generator $t^b \in T_e G$ via the adjoint map Ad_g , we expand to the linear order of α :

$$\text{Ad}_{e+i\alpha^a t^a}(t^b) = (e + i\alpha^a t^a)t^b(e - i\alpha^a t^a) \simeq t^b + i\alpha^a [t^a, t^b] + \mathcal{O}(\alpha^2) \quad (3.69)$$

This motivates us to define the infinitesimal adjoint map $\text{ad}_{t^a} \equiv d(\text{Ad})_e(t^a)$ on the Lie algebra, which acts purely as a commutator:

$$\text{ad}_{t^a}(t^b) = \lim_{\alpha^a \rightarrow 0} \frac{(\text{Ad}_{e+i\alpha^a t^a}(t^b))_{kl} - (t^b)_{kl}}{i\alpha^a} = [t^a, t^b]_{kl} = i \sum_{c=1}^d f^{abc} (t^c)_{kl}. \quad (3.70)$$

Viewing ad_{t^a} as a linear transformation matrix acting on the basis vector t^b to produce t^c , the expansion coefficients (structure constants) directly define the matrix elements of the adjoint generators:

$$(t_{\text{adj}}^a)_{bc} = i f^{abc} \quad (3.71)$$

Hence, the structure constants f^{abc} are not just algebraic constants, they are precisely **used to construct the adjoint representation**.

In this representation, the group elements act on the Lie algebra itself. The generators are $d \times d$ matrices (where $d = N^2 - 1$), with matrix elements defined by (3.71). We can verify that these matrices satisfy the Lie algebra by using the Jacobi identity:

$$\sum_{A=1}^d (f^{aAe} f^{bcA} + f^{bAe} f^{caA} + f^{cAe} f^{abA}) = 0. \quad (3.72)$$

Writing the commutation relation

$$[t_{\text{adj}}^a, t_{\text{adj}}^b]_{de} = i \sum_{c=1}^d f^{abc} (t_{\text{adj}}^c)_{de}, \quad (3.73)$$

in terms of structure constants gives:

$$\sum_{A=1}^d [(i f^{aAd})(i f^{bAe}) - (i f^{bAd})(i f^{aAe})] = i \sum_{c=1}^d f^{abc} (i f^{cde}). \quad (3.74)$$

Thus we have:

$$\sum_{A=1}^d (-f^{aAd} f^{bAe} + f^{bAd} f^{aAe}) = - \sum_{A=1}^d f^{abA} f^{Ade}. \quad (3.75)$$

Using the anti-symmetry property of structure constant: $f^{abc} = -f^{bac}$, we can reduce to the standard Jacobi identity form:

$$0 = \sum_{A=1}^d (-f^{aAd} f^{bAe} + f^{bAd} f^{aAe} + f^{Ade} f^{abA}) \quad (3.76)$$

$$= \sum_{A=1}^d (f^{aAd} f^{beA} + f^{bAd} f^{eaA} + f^{eAd} f^{abA}), \quad (3.77)$$

which is nothing but the Jacobi identity (3.72).

1.5 Inner product of generators

Now since the Lie algebra generators are all matrices, so it's natural to consider the inner product of two matrices. The most natural way is by the trace of matrix product. So we define a matrix D_r^{ab} for generators t_r^a as

$$D_r^{ab} = \text{tr}(t_r^a t_r^b). \quad (3.78)$$

For $a, b = 1, \dots, d = N^2 - 1$. Cyclicity of trace doesn't change the matrix, so it's symmetry and real-valued:

$$D_r^{ab} = D_r^{ba}, \quad (3.79)$$

and also Hermitian:

$$(D_r^{ab})^\dagger = \text{tr}((t_r^a)^\dagger (t_r^b)^\dagger) = D_r^{ab}. \quad (3.80)$$

As we know from linear algebra, for any real-valued symmetry matrix, one can always diagonalize it by a orthogonal transformation. Thus, diagonalization of D_r^{ab} gives

$$\text{tr}(t_r^a t_r^b) = C(r) \delta^{ab}. \quad (3.81)$$

Since the generators are diagonal, then the position of each components in the same t_r are equal, which imply that $C(r)$ is the same for any component of t_r . Thus $C(r)$ is called the **index of representation** or **Dynkin index**.

Example: $SU(N)$

The generators for $SU(2)$ is $t_{\text{fund}}^a = \frac{\sigma^a}{2}$ ($a = 1, 2, 3$), where σ^a is the Pauli matrix:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3.82)$$

For the fundamental representation matrix, we can check that

$$\text{tr}(t_r^a t_r^a) = \frac{1}{2}. \quad (3.83)$$

So

$$C(r) = \frac{1}{2}. \quad (3.84)$$

More over, for general case $SU(N)$, $C(r) = \frac{1}{2}$ due to the normalization of N -dim matrices are all similar with $SU(2)$. Using the index of representation, we can find the structure constants of $SU(N)$ satisfy:

$$\begin{aligned} \text{tr}([t^a, t^b] t_r^c) &= i f^{abc} \text{tr}(t_r^c t_r^c) \\ &= \text{tr}(t_r^a t_r^b t_r^c - t_r^b t_r^a t_r^c) = i f^{abc} C(r) \\ &= \text{tr}(t_r^a [t_r^b, t_r^c]) = i f^{abc} C(r). \end{aligned} \quad (3.85)$$

This shows that f^{abc} are totally anti-symmetric:

$$f^{abc} = -f^{bac} = f^{bca}. \quad (3.86)$$

As we all learned from quantum mechanics: $[J^2, J_i] = 0$. So similarly, one can define an operator T^2 that satisfies the similar commutation relation. This operator is called the **quadratic Casimir operator**, which is defined as:

$$T_{ij}^2 = \sum_{a=1}^{d(G)} \sum_{k=1}^{d(r)} t_{ik}^a t_{kj}^a, \quad (3.87)$$

where t_{ij}^a is fundamental representation. We can check that for any a , we all have: $[t^a, T^2] = 0$, namely T^2 commutes with all generators:

$$[t^a, T^2] = \left[t^a, \sum_b t^b t^b \right] = \sum_b ([t^a, t^b] t^b + t^b [t^a, t^b]) \quad (3.88)$$

$$= \sum_b \left(\left(i \sum_c f^{abc} t^c \right) t^b + t^b \left(i \sum_c f^{abc} t^c \right) \right) \quad (3.89)$$

$$= i \sum_{b,c} f^{abc} (t^c t^b + t^b t^c) \quad (3.90)$$

$$= i \sum_{c,b} f^{acb} (t^b t^c + t^c t^b) \quad (3.91)$$

$$= i \sum_{b,c} (-f^{abc}) (t^c t^b + t^b t^c) \quad (3.92)$$

$$= -i \sum_{b,c} f^{abc} (t^c t^b + t^b t^c) \quad (3.93)$$

$$= 0 \quad (3.94)$$

This implies that T^2 must be proportional to identity:

$$T_{ij}^2 = C_2(r) \delta_{ij} \quad (3.95)$$

And $C_2(r)$ is called the **quadratic Casimir coefficient**. It can be obtained by taking the trace in both sides:

$$\text{tr}(T^2) = \text{tr} \left(\sum_{a=1}^{d(G)} t^a t^a \right) = \sum_{a=1}^{d(G)} \text{tr}(t^a t^a) = C_2(r) \text{tr} \delta_{ij}, \quad (3.96)$$

thus we have:

$$C(r) d(G) = C_2(r) d(r) \quad (3.97)$$

1.6 Representation of Lorentz group

The Lorentz group, as we mentioned before, is $SO(1, 3)$ with metric $\eta = (1, -1, -1, -1)$. The group element Λ with infinitesimal expansion can be written as

$$\Lambda_\nu^\mu = \delta_\nu^\mu + i \sum_a \omega^a (t^a)_\nu^\mu + \mathcal{O}(\omega^2). \quad (3.98)$$

Defining relation of $SO(N, M)$ groups says

$$\Lambda^T \eta \Lambda = \eta, \quad (3.99)$$

thus the infinitesimal form implies

$$(\mathbb{1} + i\omega^a (t^a)^T) \cdot \eta \cdot (\mathbb{1} + i\omega^a t^a) = \eta, \quad (3.100)$$

after dropping the $\mathcal{O}(\omega^2)$, one obtains the anti-symmetric relation:

$$(t^a)^T \eta = -\eta t^a. \quad (3.101)$$

As we mentioned in the last subsection, $SO(1, 3)$, as a subgroup of $O(1, 3)$, has $\frac{1}{2} \times 4 \times (4-1) = 6$ generators, physically are 3 rotation + 3 boosts, which usually noted as $J^{\mu\nu}$. To limit the number of free parameters in $J^{\mu\nu}$, we require

$$J^{\rho\sigma} = -J^{\sigma\rho}. \quad (3.102)$$

So we change the note as

$$\sum_a \omega^a t^a \rightarrow \sum_{\rho\sigma} \omega_{\rho\sigma} J^{\rho\sigma}, \quad (3.103)$$

where $J^{\rho\sigma}$ fulfill $SO(1, 3)$ algebra. Generally, all $SO(N, M)$ groups have also spinor representations generators

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu], \quad (3.104)$$

fulfill $SO(1, 3)$ algebra:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(\eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}). \quad (3.105)$$

1.7 Integration over gauge group

Since a Lie group is also a manifold, so we can introduce the integration structure based on the group element g and function $f(g)$, the measurement dg is called Haar measure, which include the left and right invariant measure:

$$\text{left invariant measure : } \int (dg)_L f(g) = \int (dg)_L f(g'g); \quad (3.106)$$

$$\text{right invariant measure : } \int (dg)_R f(g) = \int (dg)_R f(gg'), \quad (3.107)$$

where the normalization is $\int dg \, 1 = 1$, and for most of situation the left and right Haar measure are the same $dg = (dg)_L = (dg)_R$. This measurement is just like the normal integral is invariant under the transformation $x \rightarrow x + C$, where C is a constant:

$$\int dx f(x) = \int dx f(x + C). \quad (3.108)$$

To obtain the computable form of the meature, we need explicit realization of dg with metric tensor M_{ij} :

$$M_{ij} \equiv \text{tr}(g^{-1} \partial_i g \cdot g^{-1} \partial_j g), \quad (3.109)$$

where $\partial_i g = \frac{\partial}{\partial \alpha_i} g(\alpha)$. Then we can connect to the normal integral:

$$\int dg f(g) = N \int d\alpha |\det M|^{1/2} f(g(\alpha)), \quad (3.110)$$

in which the normalization is the Jacobi determinant.

Example

- $U(1) = \{e^{i\theta} \mid -\pi < \theta < \pi\}$: For $g(\theta) = e^{i\theta}$ we have

$$M_{\theta\theta} = \text{tr}(g^{-1} \partial_\theta g \cdot g^{-1} \partial_\theta g) = \text{tr}(i \cdot i) = -1. \quad (3.111)$$

And using the fact

$$1 = N \int_{-\pi}^{\pi} d\theta \cdot 1 = N \cdot 2\pi, \quad (3.112)$$

thus $N = \frac{1}{2\pi}$. So the integral is

$$\int dg f(g) = \frac{1}{2\pi} \int_{-\pi}^{\pi} d\theta f(e^{i\theta}) \quad (3.113)$$

- $SU(2) = \{a_0 + i\vec{\sigma} \cdot \vec{a} \mid a_0^2 + \vec{a}^2 = 1\}$: For the element $g(a)$ we have:

$$\partial_0 g = \frac{\partial g}{\partial a_0} = \mathbb{1}_{2 \times 2} \quad (3.114)$$

$$\partial_k g = \frac{\partial g}{\partial a_k} = i\sigma_k \quad (k = 1, 2, 3) \quad (3.115)$$

Thus, the metric is

$$M = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}, \quad (3.116)$$

therefore, $|\det M|^{1/2} = \sqrt{16} = 4$. The last piece is the normalization factor, for that we need to compute the integral for unity. Since the constrain require a sphere, we then can express the unity by delta function:

$$1 = N \int d^4a \delta(a^2 - 1) = N \int_0^\infty r^3 dr \int d\Omega_3 \delta(r^2 - 1), \quad (3.117)$$

thus $N = \frac{1}{\pi^2}$. And finally:

$$\int dg f(g) = \frac{1}{\pi^2} \int d^4a \delta(a^2 - 1) f(g(a)). \quad (3.118)$$

- More general case, for $SU(N)$, Schur's orthogonality relation requires

$$\int dg g_{ij}(g^{-1})_{kl} = \frac{1}{N} \delta_{il} \delta_{jk}. \quad (3.119)$$

2. Quantization of non-Abelian gauge theory

2.1 Preparation

To quantize the non-Abelian gauge theory, we need to list what do we have now, everything begins with the Lagrangian:

$$\mathcal{L} = \sum_{j=1}^{d(r)} (i\bar{\psi}_j \gamma^\mu (D_\mu)_{jk} \psi_k - m\bar{\psi}_j \psi_j) - \kappa \text{tr}(F_{\mu\nu} F^{\mu\nu}), \quad (3.120)$$

where the covariant derivative is defined as:

$$(D_\mu)_{jk} = \partial_\mu \delta_{jk} - ig(A_\mu)_{jk}, \quad (3.121)$$

and the field strength is:

$$(F_{\mu\nu})_{jk} = [D_\mu, D_\nu]_{jk}. \quad (3.122)$$

Under the gauge transformation, the gauge fields obey the rule:

$$A_\mu \rightarrow A'_\mu = U A_\mu U^\dagger - \frac{i}{g} (\partial_\mu U) U^\dagger. \quad (3.123)$$

To work in the perturbation theory, we need to explicit show the colour indices. Thus we expand A_μ in basis of adjoint representation:

$$(A_\mu)_{jk} = \sum_{a=1}^{d(G)} A_\mu^a t_{jk}^a \quad (3.124)$$

The infinitesimal transformation with $U_{jk} = \mathbb{1}_{jk} + i \sum_a \alpha^a (t^a)_{jk}$ then becomes:

$$\begin{aligned} (\delta A_\mu)_{jk} &= (A'_\mu)_{jk} - (A_\mu)_{jk} \\ &= -(A_\mu)_{jk} + \left(\mathbb{1} + i \sum_a \alpha^a (t^a) \right)_{jm} (A_\mu)_{mn} \left(\mathbb{1} - i \sum_b \alpha^b (t^b) \right)_{nk} + \frac{1}{g} \sum_a (\partial_\mu \alpha^a) (t^a)_{jm} \left(\mathbb{1} - i \sum_b \alpha^b (t^b) \right)_{mk} \\ &\simeq i \sum_{a=1}^{d(G)} \alpha^a A_\mu^b [t^a, t^b]_{jk} + \frac{1}{g} \sum_{a=1}^{d(G)} (\partial_\mu \alpha^a) t_{jk}^a + \mathcal{O}(\alpha^2) \\ &= \frac{1}{g} \left[-g \sum_{b,c=1}^{d(G)} f^{abc} A_\mu^c \alpha^b + \partial_\mu \alpha^a \right] t_{jk}^a, \end{aligned} \quad (3.125)$$

thus we have

$$\delta A_\mu^a = \frac{1}{g} (D_\mu \alpha)^a = \frac{1}{g} \sum_{b=1}^{d(G)} (D_\mu)^{ab} \alpha^b. \quad (3.126)$$

where $(D_\mu)^{ab}$ is the covariant derivative in adjoint representation:

$$(D_\mu)^{ab} = \partial_\mu \delta^{ab} - g \sum_{c=1}^{d(G)} f^{abc} A_\mu^c. \quad (3.127)$$

2.2 Quantization by path integral

Now we are ready to start quantization via path integral method. We have already seen that the action $S[A]$ is gauge invariant. But

$$\int \mathcal{D}A e^{-S[A]} \quad (3.128)$$

is not well defined, because $\mathcal{D}A$ counts infinite gauge redundant. And if one integrate two times over the pure Yang-Mills term $F_{\mu\nu}^a F^{\mu\nu a}$, one can see that the momentum $M_{\mu\nu}$ satisfies

$$M_{\mu\nu} = -k^2 \eta_{\mu\nu} + k_\mu k_\nu, \quad (3.129)$$

which is degenerated (or not invertible):

$$M_{\mu\nu} k^\nu = -k^2 k_\mu + k_\mu k^2 = 0, \quad (3.130)$$

which means the determinate is 0. So we cannot define the propagator due to it's nothing but the inverse of the momentum operator, which we proved in chapter 1.

The idea to solve these problem comes from Faddeev and Popov. To divide out redundant gauge degrees of freedom, we require a gauge condition

$$G^a(A^u) = 0, \quad (3.131)$$

for a particular point A^u in an orbit of the gauge group, where u is a group element of the non-Abelian gauge group (e.g. $SU(3)$). Therefore, A^u is a gauge choice for A , in the other words, A is gauge-less. So like the way we having $\int \delta(x-a)dx = 1$, we insert the gauge condition into the delta function and define a similar quantity called **Faddeev-Popov(FP) determinant** $\Delta_G[\mathbf{A}]$:

$$\Delta_G^{-1}[A] \equiv \int \mathcal{D}u \delta(G^a(A^u)). \quad (3.132)$$

So by this definition, one can get a useful identity:

$$1 = \Delta_G[A] \int \mathcal{D}u \delta(G^a(A^u)). \quad (3.133)$$

One can check that this functional determinant is gauge invariant by continuously doing two times gauge transformation: u and u' . So the fields change to $(A^u)^{u'} = A^{u'u}$. So by definition, the new determinant is given by

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u \delta(G^a((A^{u'})^u)) = \int \mathcal{D}u \delta(G^a(A^{u'u})), \quad (3.134)$$

Now we introduce a new group element $u'' = u'u$, in which transformation the measure keeps invariant: $\mathcal{D}u = \mathcal{D}u''$, thus

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u'' \delta(G^a(A^{u''})). \quad (3.135)$$

Changing the integral variable name from u'' to u , we then verify that the FP determinant is indeed invariant under gauge transformation:

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u \delta(G^a A^u) = \Delta_G^{-1}[A] \quad (3.136)$$

Now we insert (3.133) into path integral

$$\int \mathcal{D}A (1) e^{-S[A]} = \int \mathcal{D}A \Delta_G[A] \int \mathcal{D}u \delta(G^a(A^u)) e^{-S[A]}. \quad (3.137)$$

Since only the delta function in the middle integral is u dependence, and the measure $\mathcal{D}$, the determinant $\Delta_G[A]$ and the action $S[A]$ are all gauge invariant, so we can change the variable (or choose this specific gauge) $A \rightarrow A' = A^u$ without causing any difference:

$$\int \mathcal{D}A \Delta_G[A] \int \mathcal{D}u \delta(G^a(A^u)) e^{-S[A]} = \int \mathcal{D}A'^{u^{-1}} \Delta_G[A'^{u^{-1}}] \int \mathcal{D}u \delta(G^a(A')) e^{-S[A'^{u^{-1}}]} \quad (3.138)$$

$$= \int \mathcal{D}A' \Delta_G[A'] \int \mathcal{D}u \delta(G^a(A')) e^{-S[A']}. \quad (3.139)$$

The trick that we are playing now can be understood with a simple comparison in calculus. For an integral in one dimension

$$I = \int_{-\infty}^{+\infty} dx \int_0^{2\pi} d\theta f(x + \theta)g(x + \theta), \quad (3.140)$$

which obviously is θ dependent. But we can change the variable $y \equiv x + \theta$. Since θ in the second integral is a constant, thus $dy = dx$ with the same integration range. So the integral changes to

$$I = \int_{-\infty}^{+\infty} dy \int_0^{2\pi} d\theta f(y)g(y), \quad (3.141)$$

where the integrand is now θ independent. With the same trick, now the functional integral is u independent, so we can change the integration sequence and rename the variable from A' to A as

$$\left(\int \mathcal{D}u \right) \cdot \int \mathcal{D}A \Delta_G[A] \delta(G^a(A)) e^{-S[A]}, \quad (3.142)$$

and since the integral over $\mathcal{D}u$ represents the “volume” of the gauge group, which is infinite. So we can just consider the rest part, which is physical and removed all the gauge redundancy as the correct definition of path integral for non-Abelian gauge theories:

$$\int \mathcal{D}A \Delta_G[A] \delta(G^a(A)) e^{-S[A]}. \quad (3.143)$$

So far so good, it seems that we successfully eliminated the gauge redundancy. However, if we consider a specific example, say Coulomb gauge $G^a(A) = \nabla \cdot A^a = 0$, and consider the derivative of G with respect to α , so we have

$$\frac{\delta G_\mu^a}{\delta \alpha^b} = \nabla \cdot \frac{\delta A_\mu^a}{\delta \alpha^b} = \frac{1}{g} \nabla \cdot D_\mu^{ab}, \quad (3.144)$$

where δA_μ^a is given by (3.126), and (3.127) gives

$$\vec{\nabla} \cdot \vec{D}^{ab} = \nabla^2 \delta^{ab} - g f^{abc} \vec{A}^c \cdot \vec{\nabla}, \quad (3.145)$$

which is the Coulomb gauge $\vec{\nabla} \cdot \vec{D}^{ab}$ operator. The problem is it can have non-trivial solution for A_i , namely,

$$(\vec{\nabla} \cdot \vec{D}^{ab}) \alpha^b = 0, \quad (3.146)$$

where $\alpha^b(x) \neq 0$. It means we find a new field by the gauge transformation $\alpha^b(x)$ as $A' = A + \delta A$, with the new gauge condition:

$$G^a(A') = G^a(A) + \delta G^a = 0 + \frac{1}{g} (\vec{\nabla} \cdot \vec{D}^{ab}) \alpha^b = 0, \quad (3.147)$$

which also satisfies the Coulomb gauge $\vec{\nabla} \cdot \vec{A}'^a = 0$. This problem is called the **Gribov Ambiguity**, which only troubles in non-Abelian gauge theory, e.g. QCD, due to the Abelian gauge theory have the boundary $\alpha \rightarrow 0$ at infinity and zero structure constant $f^{abc} = 0$. But for non-Abelian theory, with larger enough A in low energy confinement case, it's possible for a non-zero α satisfying

$$\nabla^2 \alpha^a = g f^{abc} \vec{A}^c \cdot \vec{\nabla} \alpha^b. \quad (3.148)$$

But at least we solved the divergent problem due to infinite gauge choice with the price of introducing a delta function and an FP determinant. These two, however also make trouble because we cannot expand perturbation series directly. The latter is limited by the delta function that only valid around $G = 0$. Therefore, to compute FP determinant, we can do the infinitesimal transformation around $G = 0$. We first consider $G(A^u)$ as function of group element $u(\alpha)$, then change the integral measure in (3.132) to: $\mathcal{D}u = \mathcal{D}G \det \left| \frac{\delta \alpha}{\delta G} \right|$, so we have:

$$\Delta_G^{-1}[A] = \int \mathcal{D}G \det \left| \frac{\delta \alpha}{\delta G} \right| \delta(G). \quad (3.149)$$

Integration over gauge group G gives the FP determinant, then we get:

$$\Delta_G[A] = \det \left| \frac{\delta G}{\delta \alpha} \right|_{G=0}. \quad (3.150)$$

So (3.143) becomes:

$$\int \mathcal{D}A \delta(G(A)) \det \left| \frac{\delta G}{\delta \alpha} \right| e^{-S[A]}. \quad (3.151)$$

Next step we are going to put $\delta(G)$ and FP determinant into the exponential to get a pure path integral form. For that, we consider the Gauss integral

$$\int_{-\infty}^{+\infty} dc e^{-\frac{c^2}{2\xi}} \delta(G - c) = e^{-\frac{G^2}{2\xi}}. \quad (3.152)$$

Generating into the functional space, with $a = 1, \dots, d(G)$, and change the gauge $G^a = 0$ to $G^a = C^a$ with constants C^a . We have:

$$\int \mathcal{D}C^a e^{-\frac{1}{2\xi} \int d^4x C^a C^a} \delta(G^a - C^a) = e^{-\frac{1}{2\xi} \int d^4x G^a G^a}, \quad (3.153)$$

where we successfully put the delta function into the exponential. The price is adding a **gauge fixing term** in Lagrangian:

$$S_{\text{gf}} = \int d^4x \frac{1}{2\xi} G^a \cdot G^a. \quad (3.154)$$

Now we left a determinant needs to be eliminated: $\det \left| \frac{\delta G}{\delta \alpha} \right|$. This can be done by introducing the Grassmann number in the path integral:

$$\det \left| \frac{\delta G^a(\alpha(x))}{\delta \alpha^b(y)} \right| = \int \mathcal{D}\bar{\eta} \mathcal{D}\eta e^{+ \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y)}, \quad (3.155)$$

where these two new field η and $\bar{\eta}$ is the **Faddeev-Popov Ghost fields**:

$$S_{\text{ghost}} = - \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y). \quad (3.156)$$

Now we are ready to write the final form of Faddeev-Popov quantization:

$$\int \mathcal{D}A \mathcal{D}\bar{\eta} \mathcal{D}\eta e^{-S[A, \bar{\eta}, \eta]}, \quad (3.157)$$

by inserting (3.154) and (3.155) into the Lagrangian:

$$S[A, \bar{\eta}, \eta] = S_{\text{YM}}[A] + S_{\text{gf}}[A] + S_{\text{ghost}}[A, \bar{\eta}, \eta] \quad (3.158)$$

$$= \int d^4x \left[\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} - \bar{\psi}(i\not{D} - m)\psi \right] + \frac{1}{2\xi} \int d^4x G^a(x) G^a(x) - \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y). \quad (3.159)$$

It is worthy to note that we used **Euclidean convention**, which is

$$\int \mathcal{D}A e^{-S}. \quad (3.160)$$

If we use **Minkowski convention** with the sign $(+ - -)$, which is

$$\int \mathcal{D}A e^{iS}. \quad (3.161)$$

So the action becomes

$$S_M[A, \bar{\eta}, \eta] = \int d^4x \left[\bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} \right] - \frac{1}{2\xi} \int d^4x G^a(x) G^a(x) + \int d^4x d^4y \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta \alpha^b(y)} \eta^b(y). \quad (3.162)$$

2.3 Feynman rules for non-Abelian gauge theory

The gauge condition we choose is the **covariant gauge (Lorenz gauge)**

$$G^a = \partial^\mu A_\mu^a = 0. \quad (3.163)$$

The quadratic terms in A_μ from $F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c$ and $G^a = \partial^\mu A_\mu^a$ can be rewritten by the partial integration as:

$$S_{\text{quadratic}} = \int d^4x \left(\frac{1}{4}(\partial^\mu A_\nu^a - \partial^\nu A_\mu^a)(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a) + \frac{1}{2\xi}(\partial^\mu A_\mu^a)(\partial^\nu A_\nu^a) \right) \quad (3.164)$$

$$= \int d^4x \left(\frac{1}{2}\partial^\mu A^{\nu a} \partial_\mu A_\nu^a - \frac{1}{2}\partial^\nu A^{\mu a} \partial_\mu A_\nu^a + \frac{1}{2\xi}(\partial^\mu A_\mu^a)(\partial^\nu A_\nu^a) \right) \quad (3.165)$$

$$= \frac{1}{2} \int d^4x A_\mu^a \left(-\partial_\rho \partial^\rho \delta_{\mu\nu} + \left(1 - \frac{1}{\xi}\right) \partial_\mu \partial_\nu \right) A^{\nu a} + \text{Boundary terms}. \quad (3.166)$$

Now with this gauge fixing term, we can check that the momentum operator becomes:

$$M_{\mu\nu} = -k^2 \eta_{\mu\nu} + \left(1 - \frac{1}{\xi}\right) k_\mu k_\nu, \quad (3.167)$$

satisfying:

$$M_{\mu\nu} k^\nu = -k^2 k_\mu + \left(1 - \frac{1}{\xi}\right) k_\mu k^2 = -\frac{1}{\xi} k^2 k_\mu \neq 0. \quad (3.168)$$

Thus, we can then define the **propagators** legally by take inverse of momentum operator, which can be read from (3.166) as

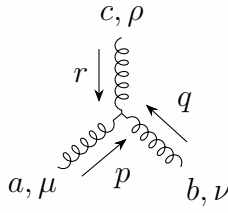
$$a, \mu \xrightarrow{p} b, \nu = \frac{-i\delta^{ab}}{p^2 + i\epsilon} \left(\eta_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right). \quad (3.169)$$

And the **gauge boson self-interaction** is vertices is given by the non-Abelian term in $\frac{1}{4}F^2$:

$$\mathcal{L}_{3A} = -gf^{abc}(\partial_\mu A_\nu^a)A^{\mu b}A^{\nu c} \quad (3.170)$$

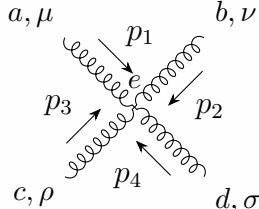
$$\mathcal{L}_{4A} = \frac{1}{4}g^2 f^{abc} f^{ade} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu}. \quad (3.171)$$

The first three gauge boson self interaction gives:



$$= -igf^{abc} [(r_\mu - q_\mu)\delta_{\nu\rho} + (q_\rho - p_\rho)\delta_{\mu\nu} + (p_\nu - r_\nu)\delta_{\mu\rho}] \quad (3.172)$$

The second four gauge boson self interaction gives:



$$= -ig^2 [f^{abe} f^{cde} (\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}) + f^{ace} f^{bde} (\delta_{\mu\nu}\delta_{\rho\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}) + f^{ade} f^{bce} (\delta_{\mu\nu}\delta_{\rho\sigma} - \delta_{\mu\rho}\delta_{\nu\sigma})]. \quad (3.173)$$

Since we have introduced a new field: ghost field $\bar{\eta}, \eta \rightarrow$, we then can also find its Feynman rule from the Lagrangian. For this, we need to first calculate the derivative by (3.126) and the gauge $G^a = \partial^\mu A_\mu^a$:

$$\frac{\delta G^a(x)}{\delta \alpha^b(y)} = \frac{\delta G^a(x)}{\delta A_\mu^c(x)} \frac{\delta A_\mu^c(x)}{\delta \alpha^b(y)} = \frac{1}{g} \frac{\delta G^a(x)}{\delta A_\mu^c(x)} D_\mu^{bc} \delta(x-y) = \frac{1}{g} \partial^\mu D_\mu^{bc} \delta(x-y), \quad (3.174)$$

with the form of covariant derivative, we have:

$$\frac{\delta G^a(x)}{\delta \alpha^b(y)} = \frac{1}{g} \partial^\mu (\partial_\mu \delta^{ab} - g f^{abc} A_\mu^c) \delta(x-y). \quad (3.175)$$

Thus the ghost action becomes:

$$S_{\text{ghost}} = \int d^4x \bar{\eta}^a(x) \left[\frac{1}{g} \partial^\mu (\partial_\mu \delta^{ab} - g f^{abc} A_\mu^c) \right] \eta^b(x) \quad (3.176)$$

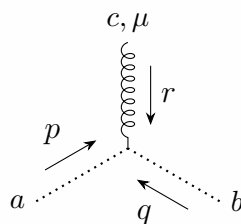
$$= \frac{1}{g} \int d^4x \bar{\eta}^a \partial_\mu \partial^\mu \eta^a - \int d^4x f^{abc} \bar{\eta}^a \partial^\mu (A_\mu^c \eta^b) \quad (3.177)$$

$$= \int d^4x \bar{\eta}^a \partial_\mu \partial^\mu \eta^a - g \int d^4x f^{abc} \bar{\eta}^a \partial^\mu (A_\mu^c \eta^b) \quad (3.178)$$

where we rescaled the field as $\eta \rightarrow \sqrt{g}\eta$ and $\bar{\eta} \rightarrow \sqrt{g}\bar{\eta}$ in the last step. Thus the first term is kinetic term and second is interaction term. So the kinetic term gives the **ghost propagators**:

$$a \xrightarrow{p} b = i \frac{\delta^{ab}}{p^2}. \quad (3.179)$$

And second interaction term gives the **vertex of ghost-ghost-gluon**:



$$= \frac{g}{2} f^{abc} (r^\mu + p^\mu - q^\mu) = -g f^{abc} q^\mu, \quad (3.180)$$

where the vertex energy-momentum conservation requires $r^\mu + p^\mu + q^\mu = 0$.

3. BRST Symmetry

3.1 Problem of negative norm

In last section, we introduced the gauge fixing term in the action to absorbing the gauge condition $G^a = 0$:

$$S_{\text{gf}} = \int d^4x \frac{1}{2\xi} G^a \cdot G^a, \quad (3.181)$$

which leads to the dynamical term of the gauge field as

$$\frac{1}{2\xi} (\partial^\mu A_\mu^a)(\partial^\nu A_\nu^a). \quad (3.182)$$

However, this operation breaks the local gauge symmetry because it contains $\partial_0 A_0$. The problem of this term is A_0 has a negative norm:

$$\langle 0 | A_\mu(k) A_\nu^\dagger(k') | 0 \rangle \propto -\eta_{\mu\nu} \delta^{(4)}(k - k'), \quad (3.183)$$

thus

$$\langle 0 | A_0(k) A_0^\dagger(k') | 0 \rangle \propto -\delta^{(4)}(k - k') < 0. \quad (3.184)$$

This should not be a problem if local gauge symmetry was not broken, because there wouldn't be the term like $\partial_0 A_0$ get into the kinematic term, i.e., propagator. So the negative norm wasn't shown when we are doing real calculations. Now everything is on the table, the optical theorem tell us we need to count all the contributions including the negative norm final states. So it may cause the sum over all physical final states (positive norm states) is greater than 1. And since the gauge choice can be changed, then the value of the probability may no longer conserved or even depends on the gauge choice, which is unacceptable.

3.2 Infinitesimal BRST transformation

To solve this problem, Becchi, Rouet, Stora and Tyutin (BRST) found that there exist a global symmetry, namely the **BRST symmetry**, which allows us to separate the non-physical states.

To introduce this symmetry, we first consider the Gaussian integral formula:

$$\int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} B^2 \right\} = \sqrt{\frac{2\pi}{a}}. \quad (3.185)$$

Change the variable $B \rightarrow B + \frac{J}{a}$:

$$\int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} \left(B + \frac{J}{a} \right)^2 \right\} = \int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} B^2 - JB - \frac{J^2}{2a} \right\} = \text{Constant}. \quad (3.186)$$

Thus, dividing $\exp\left(\frac{J^2}{2a}\right)$ on both sides, we have

$$\exp \left\{ \frac{J^2}{2a} \right\} \propto \int_{-\infty}^{\infty} dB \exp \left\{ -\frac{a}{2} B^2 - JB \right\}. \quad (3.187)$$

Now we can rewrite the gauge fixing term from the Faddeev-Popov Lagrangian by letting $J = \partial^\mu A_\mu^a$ and $a = -\xi$ as

$$\exp \left\{ i \int d^4x \left(-\frac{1}{2\xi} (\partial^\mu A_\mu^a)^2 \right) \right\} \propto \int \mathcal{D}B \exp \left\{ i \int d^4x \left(\frac{\xi}{2} (B^a)^2 + B^a (\partial^\mu A_\mu^a) \right) \right\}, \quad (3.188)$$

where B^a is an auxiliary field, which means this field is **non-dynamical**, because it doesn't contribute any term including $\partial_\mu B^a$. So the gauge fixing Lagrangian becomes:

$$\mathcal{L}_{\text{gf}} = \frac{\xi}{2}(B^a)^2 + B^a(\partial^\mu A_\mu^a). \quad (3.189)$$

We can check this Lagrangian gives the same form with the Gaussian integration by implying the Euler-Lagrange equation:

$$\frac{\delta \mathcal{L}_{\text{gauge fix}}}{\delta B^a} = \xi B^a + \partial^\mu A_\mu^a = 0, \quad (3.190)$$

$$\implies B^a = -\frac{1}{\xi} \partial^\mu A_\mu^a. \quad (3.191)$$

So the Faddeev-Popov Lagrangian is

$$\mathcal{L}_{\text{FP}} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}^a F^{\mu\nu a} + \frac{\xi}{2}(B^a)^2 + B^a(\partial^\mu A_\mu^a) + \bar{c}^a(-\partial^\mu D_\mu^{ab})c^b. \quad (3.192)$$

We have already derived variation of A from (3.126):

$$\delta A_\mu^a = D_\mu^{ab} \alpha^b(x), \quad (3.193)$$

where we absorbed the factor $1/g$ into the arbitrary function $\alpha^b(x) \rightarrow g\alpha^b(x)$. But we now want $\alpha^b(x)$ can related to the ghost field:

$$\alpha^b(x) = g\varepsilon c^b(x), \quad (3.194)$$

where ε is a constant that non-related with the space-time. So if we also do the same thing for the other fields' infinitesimal transformations, how will the Lagrangian change? This leads us to the **BRST transformations**. BRST claimed that there is a global symmetry from ε that the transformation keep the Lagrangian invariant

$$\delta_\varepsilon \mathcal{L} = 0. \quad (3.195)$$

And the infinitesimal parameter must be Grassmann-valued:

$$\varepsilon^2 = 0, \quad (3.196)$$

because A_μ^a is a bosonic field and c^b is a fermionic field, so ε must be a fermionic field, which fulfills the Grassmann algebra.

$$\varepsilon c^b = -c^b \varepsilon. \quad (3.197)$$

Thus the BRST transformation for A_μ^a and ψ_j can be derived by implying (3.194) into the normal infinitesimal transformation $\delta A_\mu^a = D_\mu^{ab} \alpha^b(x)$ and $\delta_\varepsilon \psi_j = i \sum_a \alpha^a (t^a \psi)_j$:

$$\delta_\varepsilon A_\mu^a = \varepsilon D_\mu^{ab} c^b = \varepsilon \left(\partial_\mu c^a + g \sum_{b,c} f^{abc} A_\mu^b c^c \right), \quad (3.198)$$

$$\delta_\varepsilon \psi_j = i g \varepsilon \sum_a c^a (t^a \psi)_j. \quad (3.199)$$

These two transformations obviously keep the $\mathcal{L}_{\text{YM}}$ invariant, because it just changes a name of α , and since αc^a is still a local gauge transformation, then $\mathcal{L}_{\text{YM}}$ is automatically invariant under this transformation. One can check that $F_{\mu\nu}^a$ follows

$$\delta_\varepsilon F_{\mu\nu}^a = g \varepsilon f^{abc} F_{\mu\nu}^b c^c, \quad (3.200)$$

thus

$$\delta_\varepsilon \left(-\frac{1}{4} F_{\mu\nu}^a F^{\mu\nu a} \right) = 0. \quad (3.201)$$

And for the fermionic field, the covariant derivative follows:

$$\delta_\varepsilon (D_\mu \psi) = ig\varepsilon c^a t^a (\partial_\mu \psi - igA_\mu^b t^b \psi) = ig\varepsilon c^a t^a (D_\mu \psi), \quad (3.202)$$

thus

$$\delta_\varepsilon (\bar{\psi} (i\gamma^\mu D_\mu - m) \psi) = (\delta_\varepsilon \bar{\psi}) (i\gamma^\mu D_\mu - m) \psi + \bar{\psi} (i\gamma^\mu \delta_\varepsilon (D_\mu \psi) - m \delta_\varepsilon \psi) = 0. \quad (3.203)$$

But it's worthy to mention that **BRST transformation is not a local gauge transformation (transformation parameter ε is global)**. But it indeed **comes from a local gauge transformation (transformation parameter $\alpha^a(x) = g\varepsilon c^a(x)$ is local)**.

Now it comes to the most tedious part, the gauge fixing term and the ghost term. This is because these two terms are not gauge invariant. For convenient, we can just check the following variation of the ghost field and auxiliary field:

$$\delta_\varepsilon c^a = -\frac{1}{2} g\varepsilon \sum_{b,c} f^{abc} c^b c^c \quad (3.204)$$

$$\delta_\varepsilon \bar{c}^a = \varepsilon B^a \quad (3.205)$$

$$\delta_\varepsilon B^a = 0, \quad (3.206)$$

keep the Lagrangian of gauge fixing and ghost invariant:

$$\delta_\varepsilon (\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{ghost}}) = 0, \quad (3.207)$$

namely to prove:

$$\delta_\varepsilon \left(\frac{\xi}{2} (B^a)^2 \right) + \underbrace{(\delta_\varepsilon B^a)}_{=0} (\partial^\mu A_\mu^a) + B^a \underbrace{\delta_\varepsilon (\partial^\mu A_\mu^a)}_{=\partial^\mu (\varepsilon D_\mu^{ab} c^b)} - \underbrace{(\delta_\varepsilon \bar{c}^a)}_{\varepsilon B^a} (\partial^\mu D_\mu^{ab}) c^b + \underbrace{\bar{c}^a \delta_\varepsilon (\partial^\mu D_\mu^{ab}) c^b + \bar{c}^a (\partial^\mu D_\mu^{ab}) (\delta_\varepsilon c^b)}_{\bar{c}^a \partial^\mu \delta_\varepsilon (D_\mu^{ab} c^b)} = 0. \quad (3.208)$$

The first term is simply 0 because

$$\delta_\varepsilon (B^a)^2 = 2\delta_\varepsilon B^a = 0 \quad (3.209)$$

And for the third term

$$\partial^\mu (\varepsilon D_\mu^{ab} c^b) = \underbrace{(\partial^\mu \varepsilon)}_{=0} D_\mu^{ab} c^b + \varepsilon \partial^\mu (D_\mu^{ab} c^b) = \varepsilon \partial^\mu (D_\mu^{ab} c^b), \quad (3.210)$$

due to ε is a constant. Now the expression to be proved reduces to

$$\cancel{B^a \varepsilon \partial^\mu D_\mu^{ab} c^b} - \varepsilon B^a \partial^\mu D_\mu^{ab} c^b - \bar{c}^a \partial^\mu (\delta_\varepsilon D_\mu^{ab} c^b) \quad (3.211)$$

$$= -\bar{c}^a \partial^\mu \left\{ \partial_\mu (\delta_\varepsilon c^a) + g \sum_{b,c} f^{abc} (\delta_\varepsilon A_\mu^b) c^c + g \sum_{b,c} f^{abc} A_\mu^b (\delta_\varepsilon c^c) \right\} \quad (3.212)$$

$$= -\bar{c}^a \partial^\mu \left\{ \partial_\mu \left(-\frac{g}{2} \varepsilon \sum_{b,c} f^{abc} c^b c^c \right) + g \sum_{b,c} f^{abc} \left[\varepsilon \left(\partial_\mu c^b + g \sum_{d,e} f^{bde} A_\mu^d c^e \right) \right] c^c \right. \\ \left. + g \sum_{b,c} f^{abc} A_\mu^b \left(-\frac{g}{2} \varepsilon \sum_{d,e} f^{cde} c^d c^e \right) \right\} \quad (3.213)$$

$$= -\bar{c}^a \partial^\mu \left\{ g \varepsilon \sum_{b,c} f^{abc} \left[-\frac{1}{2} (\partial_\mu c^b c^c) + (\partial_\mu c^b) c^c \right] + g^2 \varepsilon \sum_{b,c,e,f} f^{abc} \left(f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{cde} A_\mu^b c^d c^e \right) \right\} \quad (3.214)$$

The first term will be canceled by using the property of Grassmann number $c^a c^b = -c^b c^a$ and structure constant $f^{abc} = -f^{acb}$:

$$\sum_{b,c} f^{abc} \partial_\mu (c^b c^c) = \sum_{b,c} f^{abc} [(\partial_\mu c^b) c^c + c^b (\partial_\mu c^c)] \quad (3.215)$$

$$= \sum_{b,c} [f^{abc} (\partial_\mu c^b) c^c + f^{acb} c^c (\partial_\mu c^b)] \quad (3.216)$$

$$= \sum_{b,c} [f^{abc} (\partial_\mu c^b) c^c - f^{abc} c^c (\partial_\mu c^b)] \quad (3.217)$$

$$= \sum_{b,c} [f^{abc} (\partial_\mu c^b) c^c + f^{abc} (\partial_\mu c^b) c^c] \quad (3.218)$$

$$= 2 \sum_{b,c} f^{abc} (\partial_\mu c^b) c^c. \quad (3.219)$$

Now the only left term is

$$-\bar{c}^a \partial^\mu g^2 \varepsilon \sum_{b,c,e,f} f^{abc} \left(f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{cde} A_\mu^b c^d c^e \right), \quad (3.220)$$

which can be canceled by Jacobi identity

$$f^{ade} f^{bcd} + f^{bde} f^{cad} + f^{cde} f^{abd} = 0. \quad (3.221)$$

For that, we first relabel the indices d, e in the first term:

$$\sum_{b,c,d,e} f^{abc} f^{bde} c^e c^c = \frac{1}{2} \sum_{b,c,d,e} (f^{abc} f^{bde} - f^{abe} f^{bdc}) c^e c^c. \quad (3.222)$$

Thus,

$$\sum_{b,c,d,e} f^{abc} \left(f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{cde} A_\mu^b c^d c^e \right) \quad (3.223)$$

$$= \sum_{b,c,d,e} f^{abc} f^{bde} A_\mu^d c^e c^c - \frac{1}{2} f^{abc} f^{cde} A_\mu^b c^d c^e \quad (3.224)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{abc} f^{bde} - f^{abe} f^{bdc}) A_\mu^d c^e c^c - f^{abc} f^{cde} A_\mu^b c^d c^e \quad (3.225)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{ace} f^{cbd} - f^{acd} f^{cbe} - f^{abc} f^{cde}) A_\mu^b c^d c^e \quad (3.226)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (-f^{dbc} f^{ace} + f^{ebc} f^{acd} - f^{abc} f^{cde}) A_\mu^b c^d c^e \quad (3.227)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{dbc} f^{aec} - f^{ebc} f^{adc} - f^{abc} f^{dec}) A_\mu^b c^d c^e \quad (3.228)$$

$$= \frac{1}{2} \sum_{b,c,d,e} (f^{dbc} f^{aec} + f^{ebc} f^{dac} + f^{abc} f^{edc}) A_\mu^b c^d c^e \quad (3.229)$$

$$= 0. \quad (3.230)$$

Finally, with such tedious calculation, we have successfully proved (3.207) being satisfied, which means the gauge fixing term and ghost term together are BRST invariant. Thus, the Faddeev-Popov Lagrangian is BRST invariant:

$$\delta_\epsilon \mathcal{L}_{\text{FP}} = \delta_\epsilon \mathcal{L}_{\text{YM}} + \delta_\epsilon (\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{ghost}}) = 0. \quad (3.231)$$

3.3 Factorization of Hilbert space via BRST operator

Now since BRST symmetry is a global symmetry, we can in principle define a conservation charge Q , which should upgrade to an operator in Hilbert space since we have already quantized the classical field. This operator thus called the **BRST operator** Q , which can be easily defined from the infinitesimal transformation:

$$\delta_\epsilon \Phi = [\epsilon Q, \Phi] = \epsilon(Q\Phi); \quad \text{for bosonic field } \Phi \quad (3.232)$$

$$\delta_\epsilon \Psi = \{\epsilon Q, \Psi\} = -\epsilon(Q\Psi); \quad \text{for fermionic field } \Psi. \quad (3.233)$$

Thus Q has similar properties with δ_ϵ , which can be directly copy from (3.198),(3.199),(3.204),(3.205),(3.206)

$$QA_\mu^a = D_\mu^{ab} c^b \quad (3.234)$$

$$Q\psi_j = igc^a(t^a\psi_j) \quad (3.235)$$

$$Qc^a = -\frac{1}{2}gf^{abc}c^b c^c \quad (3.236)$$

$$Q\bar{c}^a = B^a \quad (3.237)$$

$$QB^a = 0, \quad (3.238)$$

and (3.196) shows that Q satisfies **Nilpotency**:

$$Q^2 = 0. \quad (3.239)$$

With BRST operator, we can rewrite the Faddeev-Popov Lagrangian as

$$\mathcal{L}_{\text{FP}} = \mathcal{L}_{\text{YM}} + Q \underbrace{\left(\bar{c}^a \partial^\mu A_\mu^a + \frac{\xi}{2} \bar{c}^a B^a \right)}_{\text{BRST ancestors}}. \quad (3.240)$$

With this form, we can easily see that it's BRST invariant by using (3.239) and $\delta_\varepsilon \mathcal{L}_{\text{YM}} = 0$:

$$Q\mathcal{L}_{\text{FP}} = 0, \quad Q\mathcal{L}_{\text{YM}} = 0 \quad (3.241)$$

Now we can use BRST operator to show how those “nonphysical” states are separated. Since BRST operator as a conserved “charge”. It naturally commutes with Hamiltonian operator H :

$$[Q, H] = 0, \quad Q = Q^\dagger. \quad (3.242)$$

The Hilbert space of eigenstates of H in Heisenberg picture can be formally expressed as

$$\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_0, \quad (3.243)$$

where the subspace can be defined by the map:

$$Q : \mathcal{H} \rightarrow \mathcal{H}. \quad (3.244)$$

The kernel of Q , i.e., those states are mapped to 0, is

$$\text{Ker}(Q) = \{|\psi\rangle \in \mathcal{H} \mid Q|\psi\rangle = 0\}, \quad (3.245)$$

and the image is

$$\text{Im}(Q) = \{|\psi\rangle \in \mathcal{H} \mid |\psi\rangle = Q|\phi\rangle, \text{ for } |\phi\rangle \in \mathcal{H}, \}. \quad (3.246)$$

And since $Q^2 = 0$, so for any states in $\text{Im}(Q)$ are all in $\text{Ker}(Q)$ too, i.e.,

$$\text{Im}(Q) \subset \text{Ker}(Q). \quad (3.247)$$

Thus, we define the three states as follows:

$$\mathcal{H}_1 = \mathcal{H} \setminus \text{Ker}(Q) = \{|\psi_1\rangle \in \mathcal{H} \mid Q|\psi_1\rangle \neq 0\}, \quad (3.248)$$

$$\mathcal{H}_2 = \text{Im}(Q) = \{|\psi_2\rangle \in \mathcal{H} \mid |\psi_2\rangle = Q|\phi\rangle, |\phi\rangle \in \mathcal{H}_1\}, \quad (3.249)$$

$$\mathcal{H}_0 = \text{Ker}(Q) \setminus \text{Im}(Q) = \{|\psi_0\rangle \in \mathcal{H} \mid Q|\psi_0\rangle = 0 \text{ and } \forall |\phi\rangle \in \mathcal{H}, |\psi\rangle \neq Q|\phi\rangle\}. \quad (3.250)$$

With these definition, we can clearly see that Hilbert space is factorized into the direct sum over three subspace :

$$\mathcal{H} = \underbrace{(\mathcal{H} \setminus \text{Ker}(Q))}_{\mathcal{H}_1} \oplus \underbrace{\text{Im}(Q)}_{\mathcal{H}_2} \oplus \underbrace{(\text{Ker}(Q) \setminus \text{Im}(Q))}_{\mathcal{H}_0}. \quad (3.251)$$

With this, we claim the set of all physical states is the kernel of $\mathcal{H}$ mode the image:

$$\mathcal{H}_{\text{phys}} \cong \mathcal{H}_0 = \text{coh}(Q) = \frac{\text{Ker}(Q)}{\text{Im}(Q)}, \quad (3.252)$$

which is a **cohomology space** satisfying:

$$Q|\psi\rangle = 0. \quad (3.253)$$

To show that it indeed make sense, we need to go back to the free theory, namely $g \rightarrow 0$, so the BRST transformation reduce to

$$QA_\mu^a = \partial_\mu c^a, \quad (3.254)$$

$$Q\bar{c}^a = B^a, \quad (3.255)$$

$$Q\psi_j = Qc^a = QB^a = 0. \quad (3.256)$$

We find this result with the definition of $\mathcal{H}_1$, $\mathcal{H}_2$ and $\mathcal{H}_0$ can help us to classify these asymptotic states:

$$A_\mu^L \in \mathcal{H}_1, \quad \bar{c}^a \in \mathcal{H}_1; \quad (3.257)$$

$$A_\mu^T \in \mathcal{H}_0, \quad \psi_j \in \mathcal{H}_0; \quad (3.258)$$

$$c^a \in \mathcal{H}_2, \quad B^a \in \mathcal{H}_2. \quad (3.259)$$

where A_μ^L represent the longitudinal polarization

$$A_\mu^L \sim k_\mu \phi \implies Q|A_\mu^L\rangle \sim |c^a\rangle \neq 0, \quad (3.260)$$

and A_μ^T represent the transversal polarization

$$k^\mu A_\mu^T = 0 \implies QA_\mu^T \sim k_\mu c^a = 0. \quad (3.261)$$

With this classification, we can see that for the states in $\mathcal{H}_1$, like A_μ^L or $\bar{c}^a$, the norm of time component are **negative**:

$$\langle A_0^L | A_0^L \rangle = \langle 0 | a_0 a_0^\dagger | 0 \rangle = \langle 0 | [a_0, a_0^\dagger] | 0 \rangle = -1 < 0, \quad (3.262)$$

$$\langle \bar{\eta} | \bar{\eta} \rangle = \langle 0 | \bar{c} \bar{c}^\dagger | 0 \rangle = \langle 0 | (-\bar{c}^\dagger)^\dagger \bar{c}^\dagger | 0 \rangle = -\langle 0 | \bar{c} \bar{c}^\dagger | 0 \rangle = \langle 0 | \{c, \bar{c}^\dagger\} | 0 \rangle = -1 < 0. \quad (3.263)$$

For the states in $\mathcal{H}_2$ are **zero norm** due to the nilpotency $Q^2 = 0$:

$$\langle \psi_2 | \psi_2 \rangle = \langle \psi_1 | Q^\dagger Q | \psi_1 \rangle = \langle \psi_1 | Q^2 | \psi_1 \rangle \equiv 0. \quad (3.264)$$

And the states in $\mathcal{H}_0$ are **positive norm**, which is the real physical state space. So we say, physical degrees of freedom are elements of $\mathcal{H}_0$, and unphysical degrees of freedom are in $\mathcal{H}_1, \mathcal{H}_2$.

This result however, one may wondering if it is picture dependent since we assume the Heisenburg picture that the states doesn't evolve with time. The answer is lucky no, picture does not change under time evolution: for any state $|\psi_0, t\rangle$ in $\mathcal{H}$, we have

$$Q|\psi_0, t\rangle = Qe^{iHt}|\psi_0\rangle = e^{iHt}Q|\psi_0\rangle = 0, \quad (3.265)$$

which means $|\psi_0, t\rangle$ must be a linear combination of states from $\mathcal{H}_0$ and $\mathcal{H}_2$, i.e., $|\psi_0, t\rangle \in \mathcal{H}_0 \oplus \mathcal{H}_2$. But if we consider a scattering process from $|\psi_0, t\rangle$ to any $|\psi_2\rangle \in \mathcal{H}_2$ that satisfies $|\psi_2\rangle = Q|\psi_1\rangle$, we have

$$\langle \psi_2 | \psi_0, t \rangle = \langle \psi_1 | Q | \psi_0, t \rangle = 0, \quad (3.266)$$

which means the scalar product vanishes! Namely, $|\psi_0, t\rangle$ has no effective component in $\mathcal{H}_2$. So even though we mix some element of $\mathcal{H}_2$ in the combination of the construction of $|\psi_0, t\rangle$, it doesn't matter because the elements in $\mathcal{H}_2$ won't contribute to the final result.

Thus, we find that **BRST operator factorize the space based on positive norm, negative norm and zero norm**. which helps us to select and drop the negative and zero norm states.